

AUTOMATIC WEATHER STATION NETWORKS

SkyGuard

A weather station reports **61.4 °C**.

Is the region on fire, or is the probe broken?

An intelligent operations and data-quality platform that answers that question, **shows its working**, estimates what the true value probably was, and tells the operator what to do — **without ever overwriting the original measurement**.

DETECT

EXPLAIN

CORRECT

ACT

Every anomaly is two different problems wearing the same clothes

FAILURE MODE A

Call real weather a fault

You "correct" a genuine heatwave out of the record. The archive now says something that never happened — and climate records are read for **decades**.

The original measurement is gone. No later improvement to the model can recover it.

FAILURE MODE B

Publish a broken probe

61.4 °C flows into every forecast, flood model, agricultural advisory and insurance calculation downstream of you.

Nobody downstream can tell. The number looks exactly like a measurement.

A platform that treats these the same is **worse than no platform at all** — it either erases real weather or ships broken data, and does so with the authority of an automated system.

Two events. Same night. Same network.

	AWS-007 • MARION RIDGE	AWS-033 • NEWPORT HEADLAND
What happened	Temperature jumps to ~61 °C in one minute	Temperature falls ~7 °C over 40 minutes
Humidity	Unchanged	Rises as the air moistens
Pressure	Unchanged	Digs a trough, then recovers
Neighbours	Nothing, anywhere in the network	The same signature — in geographic order
Rate of change	Violates the physical limit	Fast, but physically achievable
Verdict	SENSOR FAULT • 98%	GENUINE WEATHER EVENT
Action	Corrected value published, maintenance task raised	Nothing. Published as measured.

Four stages, and every one of them explains itself

STAGE 01

Detect

Seven statistical tests per channel. Z-score against a 90-minute trailing baseline, rate limits, frozen channels, variance steps, drift, range.

STAGE 02

Explain

Physics rules, neighbour comparison and propagation search. Weather or hardware — decided by weighted, named evidence.

STAGE 03

Correct

Five independent estimators combined, with a prediction interval from their disagreement. The raw value is never touched.

STAGE 04

Act

Anomalies correlate into incidents, incidents raise alerts and maintenance tasks, every decision lands in an audit log.

NO BLACK BOX

The score is a weighted sum of **named tests**, and the tests travel with it into the interface.

NO CERTAINTY

Confidence can never reach 100%. Balanced evidence returns **indeterminate**, not a coin flip.

NO DESTRUCTION

A correction is an **additional record**. The raw export still contains the fault.

Detection — measuring "unusual" in units of usual

$$z = \frac{|\text{value} - \text{baseline mean}|}{\text{baseline } \sigma}$$

z = 1 entirely ordinary

z = 2 uncommon $\approx 5\%$

z = 3 rare $\approx 0.3\%$

z = 4 the alarm point

Unit-free, so one threshold serves °C, hPa and m/s alike — and it **adapts automatically**. The same 2 °C move is alarming on a calm channel and unremarkable on a volatile one.

SPIKE / DROP

Far from its own recent average

RATE OF CHANGE

Faster than air can move

FROZEN

Bit-identical — a dead channel

NOISE • DRIFT • RANGE

Variance step, persistent offset,
impossible value

THE DETAIL THAT DECIDES WHETHER IT WORKS

A six-minute guard band

|—— baseline: 90 min ——| |guard| |
 ↑ sample under test

The baseline ends **six minutes before** the sample. A fault that ramps up would otherwise contaminate its own reference window, dragging the baseline toward the fault and shrinking the departure being measured.

This is called leakage, and it is one of the commonest ways to build a detector that quietly misses the events it was written for.

Statistics find the unusual. Physics finds the impossible.

RULE	WHAT IT CHECKS	LIMIT AS CONFIGURED
rate_of_change	Did it move faster than surface air can?	0.8 °C / min
range	Is it inside what the instrument can output?	-48 ... 56 °C
dewpoint_consistency	Dew point must not exceed air temperature	tolerance 0.6 °C
cross_sensor	Did the other channels respond as physics requires?	coherence score
neighbour	Does the station agree with its network?	3.5 σ robust
twin	Does it agree with its modelled expectation?	3.2 °C

WHY THE DISTINCTION HAS FORCE

A 5σ departure is unusual — and unusual things happen. A temperature that rose 40 °C in one minute is not unusual; there is no mechanism. **Impossible is close to proof.**

RULES THAT PASS ARE REPORTED TOO

"We checked six things, two failed, here they are" is an account an operator can **argue with**. "Anomaly detected" cannot be argued with — which means it also cannot be trusted.

Weather travels. A broken probe does not.

A station cannot audit its own thermometer — it has no second opinion.

Eleven other stations do.

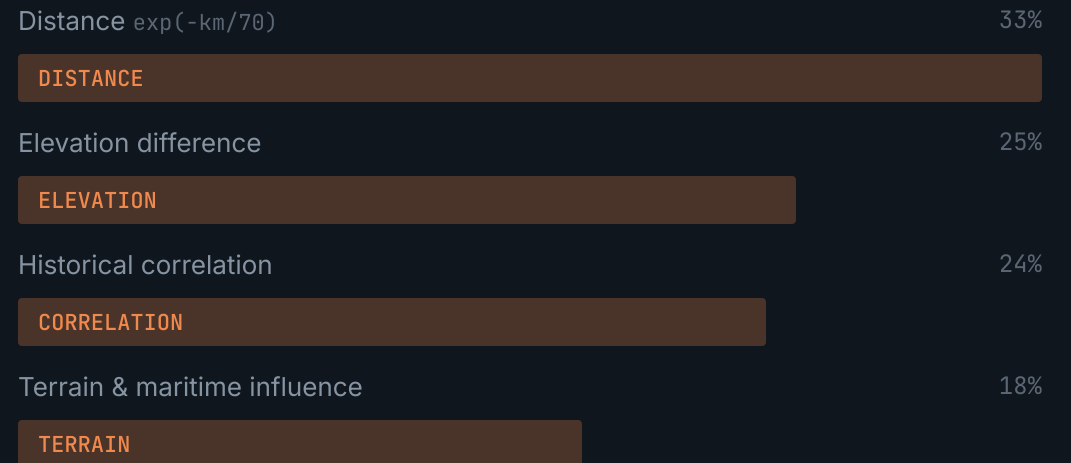
```
for each relevant neighbour:
  for lag in -180min ... +180min:
    did it move the same direction
    by at least half the magnitude?
  keep the match with smallest |lag|
```

Two matches at plausible lags, in geographic order, is strong evidence of a real event. **Zero matches — nobody, anywhere, at any lag — is strong evidence of a broken probe.**

PROXIMITY IS A POOR GUIDE

A peak 8 km away is a worse reference than a valley town 60 km away.

Relevance blends four factors:



Before any comparison: temperature is lapse-corrected (6.5 °C / 1000 m), pressure is carried via mean sea level, and humidity is converted to **dew point** and back — relative humidity is not comparable between sites at different temperatures.

A courtroom, not a formula

TOWARD HARDWARE FAULT

Physically impossible rate of change



No neighbouring station agrees



Other channels did not respond



Dew point above air temperature



Measurement variance jumped



fault 8.4 • weather 1.2

88%

Sensor fault

TOWARD GENUINE WEATHER

2.5 Same signature across the network 2.0–3.2



2.4 Every channel moved together 1.4–2.8



2.2 Digital twin tracks the station 1.5



2.0 Rate fast but achievable 1.4



1.6 Precipitation observed at the gauge 1.1



fault 1.1 • weather 6.9

86%

Weather event

fault 3.2 • weather 3.0

52%

Indeterminate — a human should look

"94% confident" is arithmetic, not a prediction

```
confidence = winning weight ÷ total weight
```

Not a probability from a model — a division over a list the operator can read line by line and disagree with. **That is the difference between a system people trust and one they eventually switch off.**

DESIGN DECISION 01

Both sides start at 0.3

A non-zero prior on the losing side mathematically caps the winner below 100%. The system is **structurally incapable** of claiming certainty about the physical world.

DESIGN DECISION 02

Below 60% → indeterminate

"I don't know, a human should look" is a **legitimate output**, and far more useful than a confident coin flip that nobody can identify afterwards.

DESIGN DECISION 03

Some findings short-circuit

A channel with **zero variance**, or a value outside what the instrument can produce, is not measuring the atmosphere at all. There is no weather hypothesis to weigh.

Five estimators — their disagreement is the error bar

SOURCE	WEIGHT	BLIND SPOT
Physics model	40%	Degrades as the fault lengthens
ML estimate (OLS)	30%	Needs quiet history to fit on
Neighbouring stations	20%	Useless if neighbours are affected too
External weather model	8%	A grid cell, not this hilltop
Recent sensor history	2%	Crude — but always available

WORKED EXAMPLE • AWS-007

raw

61.40 °C

← never modified

physics

22.10

mL

22.40

neighbours

22.00

external

21.80

history

22.30

corrected

22.16 °C

interval

21.9 – 22.5

confidence

91%

Weights are **renormalised** over whichever estimators returned a value, so a missing external model widens the interval honestly rather than quietly biasing the answer.

The rule that matters most in the entire codebase: the raw value is never overwritten. Overwriting destroys the only evidence the instrument was failing, makes the archive unauditable, and means a better model can never be applied to old data — because its input is gone.

Simulate the cause, not the symptom

STAGE 1 — THE TRUTH

One regional atmosphere

A single shared field, sampled by all twelve stations and adjusted per site for elevation, terrain and distance from the ocean.

Cold fronts, heat waves and rainfall are applied [here](#).

STAGE 2 — THE MEASUREMENT

Imperfect hardware

Each station observes that truth through real instruments: noise, calibration offsets, packet loss, battery decline.

Faults are applied [here](#), and nowhere else.

The consequence: the properties the classifier relies on are never coded — they emerge.

A front automatically appears at every station in geographic order, because they all sample the same moving field. A spike automatically appears at exactly one station, because it was added after the field was sampled. Neither behaviour is written down anywhere — which is the only reason the detection code is genuinely **under test** rather than being told the answer.

DETERMINISTIC

Nothing calls `Math.random()`. Every value is a pure function of a seed and a timestamp.

GEOMETRIC FRONTS

Arrival time is a projection onto the direction of travel, at 46 km/h.

REAL TRANSECT

Coast → Coast Range → valley → Cascades → high desert. 8 m to 1450 m.

One pure function. Fifteen screens. No two of them can disagree.

```
buildWorld({ seed, events, clock }) → World → every page reads from it
```

The usual way to build a dashboard gives each page its own fetching and its own state. It works right up until two pages disagree — and then the bug exists only in the gap between them. **SkyGuard has exactly one such gap: none.**

Inject a fault at AWS-007 and it appears on the overview, the map, the health matrix, the alert list, the maintenance queue and the audit log **in the same render** — because they are all reading the same array.

STORE INPUTS, NEVER OUTPUTS	
IN THE STORE	NEVER IN THE STORE
Clock, seed, injected events	Telemetry
The sensitivity dial	Anomalies, incidents, alerts
Acknowledgements	Health scores
Unit preferences	Anything derivable

~200ms

Full rebuild: 24 h of minute-resolution telemetry for 12 stations, plus the entire detection pipeline

17,280

Readings regenerated per rebuild — memoised against the exact inputs that produced them

0

Lines of analysis in any page component. They are pure presentation.

Six methods, scored against ground truth

"Accuracy" is a trap here. Faults occupy perhaps 40 minutes of a 1440-minute day, so a detector that reports "no fault, ever" scores 97% accurate and is completely useless.

PRECISION

When I raise an alarm, how often am I right?

Low → operators stop trusting the alerts.

RECALL

Of the real faults, how many did I catch?

Low → broken data reaches whoever depends on it.

METHOD	CHARACTER
Rule engine	Fires only on the impossible
Statistical z-score	The classic first line
Rolling deviation	Robust — MAD, not σ
Isolation Forest	32 trees, unsupervised
Digital-twin disagreement	Distance from expectation
Ensemble (production)	All of the above, with a neighbour veto

SHADOW DEPLOYMENT

v1.5-beta runs on every live request; its output goes nowhere near the operator. Weeks of real evidence, zero blast radius.

The honest caveat: these scores exist only because a simulator knows which minutes were faults. Real hardware never comes with that label — **which is precisely why the production path publishes evidence and rules rather than a score alone.** Those can be checked by a human on data that has no ground truth at all.

Fifteen sections, built for an operator on shift

MONITOR

- Network overview
- Network map
- Live monitoring

INVESTIGATE

- Anomalies
- Incidents
- Weather events
- Corrected values
- Historical analysis

ACT

- Sensor health
- Maintenance
- Alerts

SYSTEM

- Model Lab
- Reports
- Audit log
- Settings

PROVENANCE ON EVERY NUMBER

LIVE**SIMULATED****EXTERNAL****CORRECTED**

Hovering explains what it means and where the number came from. Getting this wrong would make the whole thing dishonest.

ACCESSIBILITY

Semantic landmarks, visible focus, a skip link, a keyboard-operable command palette, ARIA on every meter — and **no state signalled by colour alone**. Reduced motion honoured from the OS.

DEGRADING HONESTLY

When the providers are down the dashboard does not blank and does not invent numbers. It says so, and carries on — detection needs no network at all.

Built as production code, not a coursework demo

STACK

Framework	Next.js 16 · App Router
Language	TypeScript, strict
UI	React 19 · Tailwind CSS v4
Charts & maps	Recharts · Leaflet
State	Zustand · Zod
Live data	Open-Meteo · RainViewer (keyless)

QUALITY GATES

```
npm run lint      # ESLint + React Compiler
npm run build     # production build
npm run smoke     # headless pass over every route,
                  # failing on any console error
```

PERFORMANCE WORK WORTH NAMING

Prefix-sum rolling statistics

Rolling mean and σ over every 90-minute window in **$O(n)$** instead of $O(n \times \text{window})$ — three subtractions per query, regardless of window size.

Two-layer memoisation

React's `useMemo` plus a module-level LRU keyed on the exact inputs — correct *only* because `buildWorld` is a pure function.

Exact run-length flatness

Frozen channels are found by exact run length, not a variance threshold — prefix-sum variance of a constant series loses its significant digits to cancellation.

Eight ideas worth taking to any project

- **Show your working.** Every verdict carries the evidence that produced it.
- **Never destroy the original.** Corrections sit beside raw data, not on top of it.
- **Refuse to be certain.** No confidence reaches 100%; "indeterminate" is a valid answer.
- **One source of truth.** Screens reading the same object cannot disagree.
- **Determinism by default.** A bug you can reproduce is a bug you can fix.
- **Label everything.** Live or simulated, on every number, always.
- **Explain the threshold, not just the result.** "4.2 °C above the 90-minute baseline (7.1σ)" beats "anomaly detected".
- **Simulate the cause, not the symptom.** Then your detector is genuinely under test.

STATED PLAINLY, BECAUSE THE QUESTIONS WILL COME

The station network is **fictional** — nothing connects to real hardware. Model Lab scores exist only because a simulator knows the ground truth. Failure risk is a deterministic score, not a trained model. The operator assistant is **deliberately not a language model**: every answer is assembled from the same objects the screens render, so it cannot invent a number.

THANK YOU

SkyGuard

Detect the unusual. Explain it in evidence a human can argue with. Correct it without destroying it. Then turn it into work someone can actually do.

LIVE DEMO

skyguard-zorikos-projects.vercel.app

SOURCE

github.com/Zorik0/skyguard

PORTFOLIO

zoriko.space

Reading the code: [LEARNING_GUIDE.md](#) is a guided tour for someone handed this repository — a reading order through nine files, every concept defined once, one number traced from creation to display, and exercises. Every file in `src/lib` opens by explaining itself.